

Is ChatGPT More Energy Efficient Than My Brain?

Posted by u/oskolo • 0 points • 20 comments

Say that I assign a task to the AI that I could do myself, like checking the grammar in a text. Will the AI consume more energy than my brain would?

Comments

u/TheEqualsE • 1 points • 2023-08-24 18:39 UTC

The brain is a greedy organ in that it takes a lot of energy to run. If Chat can take that load off for you and let you spend that same energy some other way, why wouldn't you?

u/ANiceGuyOnInternet • 7 points • 2023-08-24 12:21 UTC

People are quick to say that your brain is more efficient. But that's not as clear cut as it seems.

Your brain is running a single instance of you. A server hosting GPT can be running multiple instances. Furthermore, GPT can nearly instantaneously generate text that you would take a few minutes to verbalize, possibly consuming less energy for the same thought.

So while your brain is definitely more energy efficient than, say, Intel CPUs, GPT may benefit from economies of scale.

It is hard to assess since we do not know much about the implementation of GPT4. How much computing power and storage does it need. How parallelizable is it? What architecture is it running on? For all we know it could be running extremely efficiently on an analog computer.

u/user4772842289472 • 2 points • 2023-08-24 19:10 UTC

ChatGPT is hosted on incredibly powerful hardware that the majority of us will never get anywhere near close to that kind of hardware. It draws a lot of power. It is nowhere near the efficiency of the human brain.

u/ANiceGuyOnInternet • 0 points • 2023-08-24 20:12 UTC

While it's a likely assumption, to the best of my knowledge OpenAI did not publish any information regarding the required hardware.

IBM is developing analog chips which are meant to be especially efficient for simulating neural networks. For all we know, OpenAI uses these.

u/user4772842289472 • 2 points • 2023-08-24 21:05 UTC

Even if they have cutting edge "analog chips" you have to scale up the operation so that the hardware can serve thousands of conversations happening every second through the website, add to that countless interactions that happen through the API. Conversations also happen with GPT3.5 and GPT4 at the same time. Significant hardware drawing significant power is required. You can get a sense of what is required by running your own local language model. Even 7B parameters requires cutting edge hardware that most people don't have access to. Again, scale it up to the alleged 8 models with ~200billion parameters each. The entire operation is drawing significant power beyond anything ordinary computer users can imagine.

u/ANiceGuyOnInternet • 1 points • 2023-08-24 21:32 UTC

You are probably right.

u/a_slay_nub • 1 points • 2023-08-24 12:01 UTC

If we're talking about the total energy cost per day.

Google says our brains use 12 watts of energy. This means it does $24 \times 12 = 288$ Watt Hours in a single day. GPT4 is likely run on something like 10 A/H100s and generates about 10 tokens per second. This means that it uses approximately 3-4k watts ($10 \times 250W + \text{CPU/other}$). The break-even for tokens generated per day is $288\text{WH} / 3000W \times 3600 \text{ s/H} \times 10 \text{ tokens/second} = \sim 3400$ tokens. If you are generating less than that, GPT4 is likely using less processing than your brain.

For 3.5, you could probably multiply that number by 10. Now, if you are considering full-time usage, it's not even close, GPT4 uses 3000 watts and your brain only uses 12. Someone, please check my math on this.

u/fcb0122 • 3 points • 2023-08-24 08:51 UTC

Why don't you ask ChatGPT?

u/Time_Ad_2914 • 5 points • 2023-08-24 08:46 UTC

No...? I'm sorta worried anyone would actually question this. Your brain is small as fuck compared to the massive servers that ChatGPT takes up and by the way, I read it costs 700,000 buckeroos a day to leave it operating. Not including further development costs, of course.

u/big_chestnut • 1 points • 2024-03-11 04:56 UTC

did you miss the part where chatgpt has like 100 million users?

u/DynamicHunter • 1 points • 2023-08-24 14:43 UTC

You can run a LLM on a consumer grade gaming GPU. It would be better to compare to a single instance than a server farm providing service to millions of users a day.

u/icanremembermypw4 • 4 points • 2023-08-24 11:12 UTC

You shouldn't worry, it's a legitimate and interesting question, but a difficult one to answer.

Humans are not cheap to maintain and "run" either, and we are incredibly bad/inefficient at many tasks computers and AI are good at.

I won't pretend to know the answer of the question, there are so many variables and things to

account for, you'd have to limit the question to certain parameters to even be able to calculate it, as it is impossible to account for everything "cost" wise.

u/FeltSteam • 4 points • 2023-08-24 10:18 UTC

Of course you have to factor in the reason those big data centers and inference costs are because it is generating hundreds of words a minute for tens of millions of people a month. You would definitely need a really beefy computer (And I mean beefy) but you could technically run GPT-3.5 at home (If you had the model), but the brain and body are definitely far more efficient.

u/_____Anon • 2 points • 2023-08-24 08:09 UTC

I forget the wattage but your brains like 10 watts a day or something but GPT will be 21 just for a question I imagine

u/Ur-Quan_Lord_13 • 6 points • 2023-08-24 08:35 UTC

You might mean something like "watt-hours per day" or "average wattage over the course of a day."

Brain uses about 20 watts, so over the course of a day that's 480 watt-hours.

But with current medical technology, the whole body is needed to support the brain, so really we'd want to include that in the energy cost. So it's more like 100 watts total, 2400 watt-hours per day. (Note, 2400 watt-hours is about 2000 Calories, hence the calculation; the bigger you are, the more Calories you burn, the higher that boosts the total wattage needed to power your brain.)

Ultimately that doesn't answer the question of how much energy it takes ChatGPT to answer a question, just wanted to correct the numbers and the units :p

u/_____Anon • 1 points • 2023-08-24 08:38 UTC

Op just asked if there was more or less and I put there were more used by GPT but that's interesting what you've put and sounds about right. I'm wondering now what's the most efficient brain out there for example I know dolphins have large brains but if they aren't efficient than I would say that possibly dolphins have the most inefficient brains? In terms of wattage.

I'm going to Google this now as it's quite an interesting topic.

u/Ur-Quan_Lord_13 • 1 points • 2023-08-24 08:45 UTC

Yah, just gotta multiply wattage by "time spent answering a question" to get energy used per question.

That's not that hard to figure out for the human.

To compare to ChatGPT, would need to know how much wattage is used by the few processors dedicated to the question, and for how long... Which I don't know who knows that.

u/PMMEBITCOINPLZ • 2 points • 2023-08-24 09:53 UTC

And if we're being pedantic the device used to ask the question on and all of the infrastructure usage to move that question around the internet.

u/DanTheMan_117 • 7 points • 2023-08-24 08:00 UTC

No. You need huge data centres to run chatgpt like OpenAI.

u/WithoutReason1729 • 28 points • 2023-08-24 07:39 UTC

It takes way more power for the AI to check the grammar than it does for your brain

u/badhoccyr • 1 points • 2023-10-23 17:47 UTC

But it does it much faster

u/AutoModerator • 1 points • 2023-08-24 06:35 UTC

Hey /u/oskolo, if your post is a ChatGPT conversation screenshot, please reply with the [conversation link](<https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq>) or prompt. Thanks!

We have a [public discord server](<https://discord.gg/r-chatgpt-1050422060352024636>). There's a free Chatgpt bot, Open Assistant bot (Open-source model), AI image generator bot, Perplexity AI bot, 🤖 GPT-4 bot ([Now with Visual capabilities (cloud vision)!](<https://cdn.discordapp.com/attachments/812770754025488386/1095397431404920902/image0.jpg>)) and channel for latest prompts! New Addition: Adobe Firefly bot and Eleven Labs cloning bot! [So why not join us?](<https://discord.com/servers/1050422060352024636>))

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PSA: For any Chatgpt-related issues email support@openai.com

I am a bot, and this action was performed automatically. Please [contact the moderators of this subreddit](/message/compose/?to=r/ChatGPT) if you have any questions or concerns.